Please write clearly in block capitals. Centre number | Candidate number

Surname

Forename(s)

A-level

ENVIRONMENTAL SCIENCE

Paper 1 (Generated practice paper - Set 27)

Assessed: The physical environment, Energy resources, Pollution, Research methods

Time allowed: 3 hours

Materials

For this paper you may use:
- a calculator.

Instructions

- Use black ink or black ball-point pen. Pencil should only be used for drawing.
- Fill in the boxes at the top of this page.
- Answer **all** questions 1 to 10 and **one** essay from question 11.
- You must answer the questions in the spaces provided. Do not write outside the box around each page or on blank pages.
- Do all rough work in this book. Cross through any work you do not want to be marked.

Information

- The marks for questions are shown in brackets.
- The maximum mark for this paper is 120.
- All questions should be answered in continuous prose.
- You will be assessed on your ability to:
  - use good English
  - organise information clearly
  - use specialist vocabulary where appropriate.

For Examiner's Use

Question	Mark
1	
2	
3	
4	
5	
6	
7	
8	
9	
10	
11	
TOTAL	

Do not write outside the box

Answer all questions in the spaces provided.

0 1 . 1 Which of the following is a feature of a river that affects its usefulness as a water source? Tick **one** box.

[1 mark]

A Its total annual discharge ☐

B The albedo of its surface ☐

C Its salinity ☐

D The age of its bedrock ☐

0 1 . 2 Explain why rainwater collection is increasingly important in cities such as those in India.

[2 marks]

.....

.....

.....

.....

0 1 . 3 Explain why an estuary barrage has fewer land-use conflicts than a reservoir on land but still causes environmental problems.

[2 marks]

.....

.....

.....

.....

5

0 2 . 1 Which of the following best describes the purpose of Critical Group Monitoring? Tick one box.

[1 mark]

A To detect health damage in the population most affected by a pollutant ☐

B To predict the route a pollutant will take through the environment ☐

C To assess exposure of the people most at risk so that everyone else can be assumed to be sa ☐

D To measure the concentration of a pollutant at the point of release ☐

0 2 . 2 Explain why CPA can reduce the cost of an environmental monitoring programme.

[2 marks]

Do not write outside the box

Do not write outside the box

.....

.....

.....

.....

0 2 . 3 A new chemical has properties similar to PCBs (persistent, liposoluble chlorinated organic compounds). Explain how knowledge of PCBs helps decisions about the new chemical.

[2 marks]

.....

.....

.....

.....

5

0 3 A student measured the mass of carbon stored per hectare in three land uses in the same region.

Table 1

Land use	Carbon in vegetation / t ha^{-1}	Carbon in soil dead organic matter / t ha^{-1}
Mature woodland	120	95
Permanent pasture	3	80
Arable (ploughed annually)	2	45

0 3 . 1 Calculate the total carbon lost per hectare when mature woodland is converted to arable farmland.

[1 mark]

Answer _____ t/ha

0 3 . 2 Explain why the soil carbon in arable land is so much lower than in pasture, even though the vegetation carbon is similar.

[2 marks]

.....

.....

.....

.....

0 3 . 3 Suggest **two** ways the arable farmer could increase soil carbon storage.

[2 marks]

1 .....

.....

2 .....

.....

5

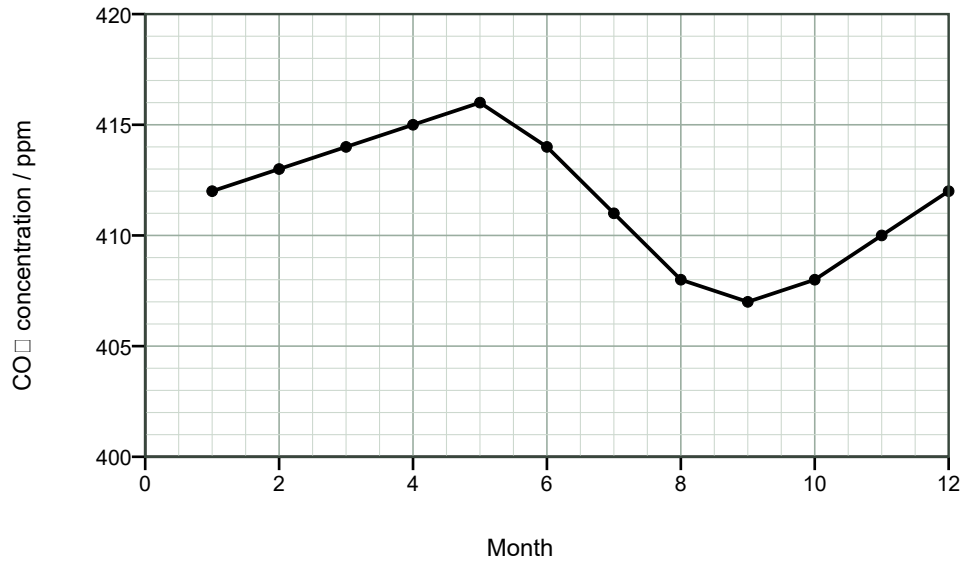
Do not write outside the box

0 4 Table 1 shows the composition of the dry atmosphere. Figure 1 shows the changes in carbon dioxide concentration measured at a site in the northern hemisphere over one year.

Table 1

Gas	Percentage by volume / %
Nitrogen	78
Oxygen	21
Argon and other rare gases	0.9
Carbon dioxide	0.04
Ozone	0.000007

Figure 1



0 4 . 1 Describe and explain the pattern shown in Figure 1.

[4 marks]

.....

.....

.....

.....

.....

.....

.....

.....

.....

Do not write outside the box

0 4 . 2 Explain why the atmosphere is described as being in a state of **dynamic equilibrium**.

[2 marks]

.....

.....

.....

.....

0 4 . 3 Give **two** ways in which the atmosphere supports life other than by providing gases for biological molecules.

[2 marks]

1 .....

.....

2 .....

.....

0 4 . 4 Name the two layers of the atmosphere most affected by human activities.

[2 marks]

1 .....

.....

2 .....

.....

10

0 5 Table 1 shows the annual mean concentrations of four pollutants measured at a city-centre monitoring station, with the legal limit for each.

Table 1

Pollutant	Annual mean / µg m^{-3}	Legal limit / µg m^{-3}
Nitrogen dioxide	52	40
PM10 particulates	28	40
PM2.5 particulates	19	20
Sulfur dioxide	8	125

0 5 . 1 Calculate by what percentage the annual mean nitrogen dioxide concentration exceeds the legal limit.

[2 marks]

Do not write outside the box

Do not write outside the box

Answer [ ] %

0 5 . 2 Suggest why the sulfur dioxide concentration is now far below its limit while nitrogen dioxide is not.

[3 marks]

.....

.....

.....

.....

.....

.....

.....

0 5 . 3 Explain how each of the following reduces emissions.

[3 marks]

Electrostatic precipitator .....

.....

Catalytic converter .....

.....

Low-NOx burner .....

.....

0 5 . 4 Explain why a reduction in the city's NOx emissions would also reduce summer ground-level ozone.

[3 marks]

.....

.....

.....

.....

Do not write outside the box

0 5 . 5 The council also plans to monitor air quality using lichens. Explain the advantages and the limitations of using lichens as indicator species.

[4 marks]

15

0 6 . 1 Explain how each of the following human activities increases soil erosion.

[4 marks]

Overgrazing

Reducing soil biota

0 6 . 2 Explain why landslides become more likely on steep slopes some years after deforestation.

[2 marks]

Do not write outside the box

.....

.....

.....

0 6 . 3 Describe **four** environmental impacts of accelerated soil erosion away from the field where it occurs.

[4 marks]

1 .....

.....

2 .....

.....

3 .....

.....

4 .....

.....

10

0 7 . 1 Explain why compressed air energy storage with heat recovery can achieve over 90% efficiency.

[2 marks]

.....

.....

.....

.....

0 7 . 2 Explain why large-volume hot water stores lose heat more slowly per unit of stored energy than small ones.

[2 marks]

.....

.....

.....

.....

0 7 . 3 Explain why molten salt rather than water is used to store heat from CSP plants.

[2 marks]

.....

.....

.....

.....

0 7 . 4 Explain how high thermal mass building materials reduce energy use.

[2 marks]

.....

.....

.....

.....

0 7 . 5 Describe **one** other energy storage technology.

[2 marks]

.....

.....

.....

.....

10

0 8 Table 1 shows the volume of water in some reservoirs of the hydrological cycle and the annual transfer rates through them.

Table 1

Reservoir	**Volume / x 10^{15} dm^3**	**Annual transfer (inflow = outflow) / x 10^{15} dm^3 yr^{-1}**	**Percentage of global water / %**
Oceans	1 340 000	425	97
Ice caps and glaciers	24 000	3	1.7
Groundwater	23 000	12	1.7
Lakes and rivers	180	40	0.013
Soil moisture	17	80	0.001
Atmosphere	13	480	0.001

Do not write outside the box

0 8 . 1 The residence time of water in a reservoir is given by: residence time = volume in reservoir / annual transfer rate. Use Table 1 to calculate the residence time of water in the atmosphere. Give your answer to the nearest whole day.

[3 marks]

Answer ☐ days

0 8 . 2 Calculate the residence time of water in the oceans.

[1 mark]

Answer ☐ years

0 8 . 3 Explain why the atmosphere is an important reservoir even though it holds only 0.001% of global water.

[2 marks]

.....

.....

.....

.....

0 8 . 4 Explain why human changes to a small reservoir such as soil moisture are more obvious than changes to the oceans.

[2 marks]

.....

.....

.....

Do not write outside the box

.....

0 8 . 5 Explain why the hydrological cycle is described as being in dynamic equilibrium.

[2 marks]

.....

.....

.....

.....

10

0 9 . 1 Explain what is meant by the **embodied energy** of a vehicle and how it can be reduced.

[3 marks]

.....

.....

.....

.....

.....

.....

.....

0 9 . 2 Describe **four** principles of designing vehicles for end of life.

[4 marks]

1 .....

.....

2 .....

.....

3 .....

.....

4 .....

.....

Do not write outside the box

0 9 . 3 Explain why there is an optimum speed for fuel efficiency in a car.

[2 marks]

.....

.....

.....

.....

0 9 . 4 Give **one** way driver behaviour or vehicle systems can reduce fuel use in congested traffic.

[1 mark]

.....

.....

10

1 0 A student investigated whether the number of earthworms per m^2 differs between an organic field and a conventionally farmed field. Table 1 shows the results.

Table 1

Field	Number of samples	Mean earthworms per m^2	Standard deviation
Organic	10	48	14
Conventional	10	31	12

1 0 . 1 Name the statistical test the student should use and explain why it is appropriate.

[2 marks]

.....

.....

.....

.....

1 0 . 2 The test gave p = 0.01. State the conclusion the student should draw.

[2 marks]

.....

.....

.....

.....

Do not write outside the box

1 0 . 3 Explain why the student cannot conclude that organic farming **caused** the higher earthworm numbers.

[2 marks]

1 0 . 4 Evaluate the decisions that must be made when planning a fieldwork investigation so that valid conclusions can be drawn.

[9 marks]

Do not write outside the box

Do not write outside the box

Extra space

15

Write an essay on one of the following topics.

1 1 . 1 Discuss how an understanding of the hydrological cycle can be used to develop strategies for the sustainable management of water resources.

[25 marks]

OR

1 1 . 2 Discuss how knowledge of the properties of pesticides can be used to reduce their environmental impacts while maintaining food production.

[25 marks]

Indicate which question you have answered by writing its number, **11 . 1** or **11 . 2**, at the top of your answer.

Do not write outside the box

Do not write outside the box

Do not write outside the box

Do not write outside the box

Do not write outside the box

END OF QUESTIONS